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Experiment - 1
Caesar cipher

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Aim:- To implement Caesar cipher encryption using a key.

Algorithm:-

1. Read plaintext and key.
2. Shift each letter by the keyvalue.
3. Display the ciphertext.

Pseudocode:-

```
BEGIN
    GET plaintext and key
    FOR each letter in phintext
        FIND shifted letter using key
        STORE shifted letter
    END FOR
    PRINT ciphertext
END
```

INPUT:-

TEXT = HELLO, Key = 3

OUTPUT:-

KHOOR

Pseudocode:-

```
key = 3
for (i=0; text[i]!='\0'; i++)
        cipher[i] = (text[i] - 'A' + key) % 26 + 'A';
cipher[i] = '\0';
```

Experiment-2
Monoalphabetic Substitution Cipher

Aim:-

To implement monoalphabetic substitution cipher encryption.

Algorithm:-

1. Read plaintext and substitution key
2. Replace each letter using the key.
3. Display the ciphertext.

Pseudocode:-

```
BEGIN
    GET plaintext and key
    FOR each letter
        FIND its substitute using key
        STORE substituted letter
    END FOR
    PRINT ciphertext
END
```

INPUT:-

plaintext = ABC ; key = QWERTYIOUPASDFGHJKLZXCVBNM

OUTPUT:-

QWE

Pseudocode:-

```
key = "QWERTYIOUPASDFGHJKLZXCVBNM";
for(i=0;text[i]!='\0';i++)
        cipher[i] = key[text[i]-'A'];
cipher[i]='\0';
```

Experiment - 3
Playfair cipher

Aim:-

To implement playfair cipher encryption using a 5×5 matrix.

Algorithm:-

1. Read a plaintext and keyword.
2. Create a 5×5 matrix using the keyword.
3. Encrypt letters in pairs using matrix rules.
4. Display the ciphertext.

Pseudocode:-

```
BEGIN
    GET plaintext and keyword
    CREATE 5x5 key matrix
    SPLIT plaintext into pairs
    Encrypt each pair using matrix rules.
    PRINT ciphertext
END
```

Input:-

Plaintext = HELLO , Key = MONARCHY

OUTPUT:-

Ciphertext = CFSUPM.

Pseudocode:-

```
key = 'MONARCHY'
create_matric(key)
for (i=0; i<n; i+=2)
    encrypt_pair(text[i], text[i+1]);
cipher[i] = '\0';
```

Experiment - 4
polyalphabetic substitution Cipher

Aim:- To implement polyalphabetic substitution cipher encryption using a key.

Algorithm:-

1) Read the plaintext and keyword.
2) Repeat the keyword until its length matches the plaintext.
3) Convert each plaintext and key letter into a number from 0 to 25.
4) Add the corresponding values and apply modulo 26.
5) Convert the result into ciphertext letters.
6) Display the encrypted message.

Pseudocode:-

```
key = "key"
for(i=0; i<n; i++)
    k[i] = key[i%strlen(key)];
for(i=0; i<n; i++)
    cipher[i] = (text[i]-'A' + k[i]-'A' %26+'A';
Cipher[n]='\0';
```

Input:-

plaintext = HELLO, KEY = KEY

OUTPUT:-

RIJVS

Experiment - 5
Affine Caesar Cipher

Aim:- To implement the Affine Caesar cipher using encryption formula.

Algorithm:-

1. Read the plaintext and keys a and b
2. Convert each plaintext letter into a number from 0 to 25.
3. Calculate the ciphertext using $C = (ap+b) \bmod 26$
4. Convert the result back into a letter
5. Display the ciphertext and ensure that a is coprime with 26

Pseudocode:-

```
Scanf("%s%d%d", text, &a, &b);
for(i=0; text[i]!='\0'; i++){
        p = text[i] - 'A';
        cipher[i] = ((a * p + b) % 26) + 'A';
}
printf("%s", cipher);
```

Input:-

Text = HELLO, a = 5, b = 8

Output:-

RCLLA

Experiment-6
Breaking the Affine cipher

Aim:- To implement a C program to break an affine cipher using letter frequency analysis.

Algorithm:-

1. Read the encrypted ciphertext
2. Count the frequency of each letter in the ciphertext
3. Identify the most frequent letter, B, and, the second most frequent letter U.
4. Assume these letter may represent common English letters, such as E and T.
5. Use the affine encryption formula $C=(ap+b) \bmod 26$ to determine possible keys.
6. Try valid key combinations where a is coprime with 26.
7. Decrypt the ciphertext using each key and identify the meaningful plaintext.

Pseudocode:-

```
fscanf("1.s", cipher);
for (i = 0; cipher[i] != '\0'; i++)
    freq[cipher[i] - 'A']++;
Find most frequent (freq);
for (a=1; a<26; a++)
    try affine decryption(cipher, a);
```

Input:-

Ciphertext = BUBU

Output:-

Most frequent letter = B
Second most frequent letter = U
possible plaintext: ETET

Experiment-7

Breaking a Monoalphabetic substitution cipher

Aim:- To implement a C program to decrypt a monoalphabetic substitution cipher using letter frequency analysis.

Algorithm:-

1. Read the given ciphertext.
2. Count the frequency of each symbol in the cipher.
3. Identify the most frequently occuring symbol
4. Assume the common symbols represent frequent English letters such as E, T and A.
5. Analyze repeated symbols and common word patterns.
6. Substitute the guessed letters to form possible way.
7. Display the meaningful decrypted message.

Pseudocode:-

```
&scanf("%s", cipher);
for (i=0; cipher[i]!='\0'; i++;
    freq[cipher[i]]++;
find common symbols(freq);
for (i=0; i<strlen(cipher); i++;
    decrypt using mapping(cipher[i]);
printf("%s", plaintext);
```

Input:-

Ciphertext = BQQ

Output:-

Possible plaintext = ALL